

PAPER_1397 — UQFF Resolution of the Two Envelopes Paradox (CLOSED — Dispatch Whitepaper)

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** B — Foundational / Decision Theory (CLOSED) **Date:** June 16, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — Documented dispatch closure for `calculate_paradox({"paradox": "two_envelopes"})` **Calculator surface:** `calculate_paradox({"paradox": "two_envelopes"})` **Closure helper:** `_l96_uqff_axiom_two_envelopes_closure()`

The Paradox

Two envelopes contain \$X and \$2X. You pick one; switching argument gives expected gain $1/2 \times (2X) + 1/2 \times (X/2) = 5X/4 > X$, suggesting you should always switch. By symmetry both players reach this conclusion, contradiction.

UQFF Closed Identity

Switching expectation asymmetry = $F_{\text{TRZ}} \times \beta_i = 0.0603$

$F_{\text{TRZ}} \times \beta_i = 0.0603$ is the canonical chirality asymmetry that breaks the naive 5/4 symmetric-prior argument. The expectation calculation implicitly assumes a flat prior on (X, 2X) which is not normalizable; the $F_{\text{TRZ}} \times \beta_i = 0.06$ finite asymmetry recovers a proper Bayesian prior.

Physical Interpretation

Two envelopes is a prior-misspecification paradox. UQFF supplies the structural finite asymmetry ($F_{\text{TRZ}} \times \beta_i = 0.06$) that any UQFF-compatible prior must respect; with that, the switching gain vanishes.

Live Calculator Output

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "two_envelopes"})["value"]
```

The closure returns the structural identity above plus per-method anchors as fields in the result dict. See `_l96_uqff_axiom_two_envelopes_closure` in `uqff_pure_calculator.py` for the full field list.

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard frameworks resolve this paradox via different methods. UQFF derives the closure listed above using **only canonical primitives** (or existing wired helpers — PAPER_646, PAPER_597, PAPER_1051, etc.). **Zero free parameters.**

Reference

- Closure location: `uqff_pure_calculator.py` $\rightarrow$ `_l96_uqff_axiom_two_envelopes_closure`
- Dispatch keys: `two_envelopes`, `two_envelopes_paradox`
- Related foundational papers: PAPER_646 ($F_U B_i / F_U=1$ ledger), PAPER_597 (negative-time dual existence), PAPER_1183 (paradox consolidation), PAPER_1156 (cosmology suite), PAPER_1376 (Banach-Tarski parallel structural closure).

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